

CPRI Hackathon Screening Round

Methodology Note - Automated Validity Screening and Reference Prediction

1. Objective

The program processes historical and new laboratory tests without editing individual records. It identifies unreliable measurements, predicts the verified hot-spot temperature-rise parameter for every test, and automatically creates the required summary and attention list.

2. Dataset audit and important parameters

Measure	Finding	Use in method
Training records	1,000	866 Valid; 134 Invalid
Test records	350	All require a prediction and label
Exact duplicate rows	24 train; 8 test	Deterministic logging-error signal
Primary sensor gaps	6 test rows	Invalid; imputed only for regression
S4 gaps	11 test rows	Not a validity fault

Applied voltage, load current, ambient temperature and duration define the operating state. S1-S3 contain consistent thermal information in Valid tests. S4 is excluded from the model: it is nearly uncorrelated with the reference and its missing values occur in Valid records.

3. Abnormal-record identification

1. Flag exact duplicate measurement records as Invalid.
2. Flag a missing S1, S2 or S3 measurement as Invalid. A missing S4 alone is allowed.
3. Fit three linear response models on historical Valid tests to estimate expected S1-S3 from the operating conditions. The anomaly score is the largest absolute sensor residual.
4. Calibrate the residual threshold from labelled history. The maximum Valid score was 1.044, while the minimum non-structural fault score was 4.227. The selected midpoint threshold is 2.635.

This conditional sensor check protects genuine extreme operating regimes: a high temperature or current is not rejected when all primary sensors remain physically consistent.

4. Reference Parameter prediction

Only engineer-labelled Valid rows train the regressor. The model uses voltage, current, ambient temperature, duration and S1-S3, with polynomial thermal terms and explicit regime features. Cross-validation located a sharp load transition near 62 A and a high-ambient slope change. Five nearby breakpoint models are averaged to reduce threshold sensitivity. Missing or clearly faulty primary sensors are replaced by their operating-condition estimates before prediction.

Validation, automation and digital-twin deployment

5. Leakage-safe validation

Ten-fold validation was used. Exact duplicates were kept in the same classification fold. All preprocessing, sensor-response fitting and threshold calibration were repeated inside each fold. Regression metrics were calculated only on historical Valid records because Invalid rows can contain irrecoverable logging corruption.

Task	Metric	Out-of-fold result
Validity	Accuracy / balanced accuracy	1.000 / 1.000
Validity	Invalid precision / recall / F1	1.000 / 1.000 / 1.000
Reference	MAE / RMSE	0.1935 C / 0.2429 C
Reference	R-squared / maximum absolute error	0.999488 / 0.9046 C

Perfect historical classification is reported as an out-of-fold result, not training accuracy. It supports this dataset's clean separation but does not prove that future unseen faults are always detectable. The regression residual is consistent with remaining measurement noise.

6. Automated outputs

- Team CSV: Test_ID, predicted Reference Parameter and Validity Label.
- JSON summary: record count, invalid count, minimum, maximum, average and top three attention IDs.
- Validation JSON: audit counts, thresholds, cross-validation metrics and integrity checks.

The attention rank combines invalid status, sensor-residual severity, primary-sensor gaps, duplicate status, repaired-sensor count and prediction disagreement.

7. Assumptions and limitations

Operating-condition fields are assumed present and correctly scaled. Exact duplicates are treated as logging errors because all such historical rows are Invalid. S4 remains auxiliary. A predicted reference for an Invalid record is an estimate after sensor repair; a duplicated record may still contain irrecoverable target ambiguity. No method can guarantee zero error on future noisy data.

8. Digital-twin automation steps

1. Ingest each test directly from the acquisition system with schema, unit and ID validation.
2. Store raw records immutably and calculate duplicate, missingness and sensor-residual checks.
3. Apply the Valid/Invalid gate and route Invalid/high-attention tests to engineer review.
4. Repair only clearly unreliable sensor values; retain the original values and reason codes.
5. Run the regime-aware reference model and publish prediction plus uncertainty indicators.
6. Monitor drift in operating ranges, residuals, invalid rate and prediction errors after calibration checks.
7. Version data, thresholds and models; retrain only after approved reference measurements accumulate.